

ADVANCED

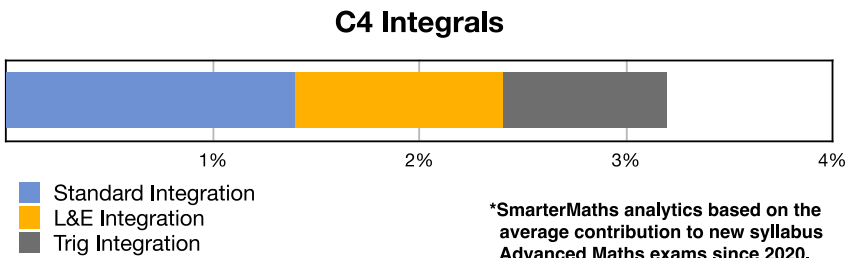
Stage 6

Calculus (Adv), C4 Integration (Adv), Integration (Y12)

L&E Integration

Teacher: SHS Maths

Exam Equivalent Time: 79.5 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: *1-Integrals* (3.2%), *2-Areas Under Curves* (5.8%), *3-Trapezoidal Rule* (2.2%) and *4-Other Integration Examples* (2.8%).
- This analysis looks at *Integrals*.

HSC ANALYSIS - What to expect and common pitfalls

- Integrals* (3.2%) is made up of standard integration, log and exponential integration and trig integration.
- Dedicated questions testing log and exponential integration have regularly appeared in past HSC exams (notably absent in 2021-24), including 3 separate questions in the 2020 exam.
- Questions requiring students to differentiate then integrate have easily been the worst answered, causing problems in 2020 and 2019 - an important revision area.
- Integration of a fraction that results in a logarithm is a 2-3 mark question type has a strong history of being examined, but noticeably just once in the last 6 years (2020 Adv 17).
- We note 2019 13c(ii) surprisingly produced a sub-50% mean mark and deserves attention.

Questions

1. Calculus, 2ADV C4 2014 HSC 4 MC

Which expression is equal to $\int e^{2x} dx$?

- A. $e^{2x} + C$
- B. $2e^{2x} + C$
- C. $\frac{e^{2x}}{2} + C$
- D. $\frac{e^{2x+1}}{2x+1} + C$

2. Calculus, 2ADV C4 2020 HSC 4 MC

What is $\int e + e^{3x} dx$?

- A. $ex + 3e^{3x} + c$
- B. $ex + \frac{1}{3}e^{3x} + c$
- C. $e + 3e^{3x} + c$
- D. $e + \frac{1}{3}e^{3x} + c$

3. Calculus, 2ADV C4 2012 HSC 9 MC

What is the value of $\int_1^4 \frac{1}{3x} dx$?

- A. $\frac{1}{3}\ln 3$
- B. $\frac{1}{3}\ln 4$
- C. $\ln 9$
- D. $\ln 12$

4. Calculus, 2ADV C4 2013 HSC 11e

Find $\int e^{4x+1} dx$ (2 marks)

5. Calculus, 2ADV C4 2004 HSC 3bi

Evaluate $\int_1^2 e^{3x} dx$. (2 marks)

6. Calculus, 2ADV C4 2006 HSC 2bi

Find $\int 1 + e^{7x} \, dx$. *(2 marks)*

7. Calculus, 2ADV C4 2015 HSC 11h

Find $\int \frac{x}{x^2 - 3} \, dx$. *(2 marks)*

8. Calculus, 2ADV C4 2018 HSC 11e

Evaluate $\int_0^3 e^{5x} \, dx$. *(2 marks)*

9. Calculus, 2ADV C4 2016 HSC 12d

- i. Differentiate $y = xe^{3x}$. *(1 mark)*
- ii. Hence find the exact value of $\int_0^2 e^{3x}(3 + 9x) \, dx$. *(2 marks)*
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10. Calculus, 2ADV C4 2004 HSC 3bii

Find $\int \frac{x}{x^2 - 3} \, dx$. *(2 marks)*

11. Calculus, 2ADV C4 2006 HSC 2bii

Evaluate $\int_0^3 \frac{8x}{1 + x^2} \, dx$. *(3 marks)*

12. Calculus, 2ADV C4 2019 MET1-N 3

Evaluate $\int_0^1 e^x - e^{-x} \, dx$. *(2 marks)*

13. Calculus, 2ADV C4 2020 HSC 17

Find $\int \frac{x}{4 + x^2} \, dx$. *(2 marks)*

14. Calculus, 2ADV C4 SM-Bank 1

If $m = \int_1^3 \frac{2}{x} \, dx$, express e^m in its simplest form. *(2 marks)*

15. Calculus, 2ADV C4 SM-Bank 3

If $f'(x) = \frac{x^2}{x^3 + 1}$ and $f(1) = \log_e 2$, find $f(x)$. *(3 marks)*

16. Calculus, 2ADV C4 2005 HSC 2ci

Find $\int \frac{6x^2}{x^3 + 1} \, dx$. *(2 marks)*

17. Calculus, 2ADV C4 2008 HSC 2ci

Find $\int \frac{dx}{x + 5}$. *(1 mark)*

18. Calculus, 2ADV C4 2010 HSC 2dii

Find $\int \frac{x}{4 + x^2} \, dx$. *(2 marks)*

19. Calculus, 2ADV C4 2011 HSC 4b

Evaluate $\int_e^{e^3} \frac{5}{x} \, dx$. *(2 marks)*

20. Calculus, 2ADV C4 2012 HSC 12b

Find $\int \frac{4x}{x^2 + 6} \, dx$. *(2 marks)*

21. Calculus, 2ADV C4 2013 HSC 11f

Evaluate $\int_0^1 \frac{x^2}{x^3 + 1} \, dx$. *(3 marks)*

22. Calculus, 2ADV C4 2019 HSC 13c

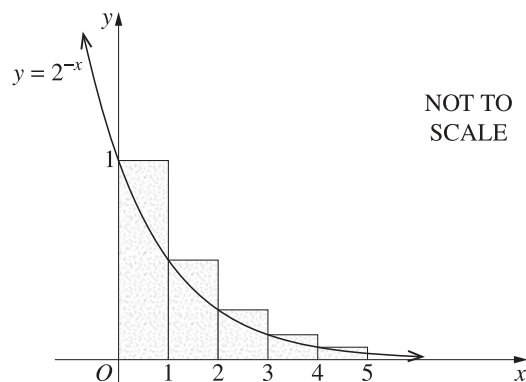
- i. Differentiate $(\ln x)^2$. *(2 marks)*
- ii. Hence, or otherwise, find $\int \frac{\ln x}{x} \, dx$. *(1 mark)*
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23. Calculus, 2ADV C4 2020 HSC 18

- a. Differentiate $e^{2x}(2x + 1)$. *(2 marks)*
- b. Hence, find $\int (x + 1)e^{2x} \, dx$. *(1 marks)*
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24. Calculus, 2ADV C4 2022 HSC 29

- a. The diagram shows the graph of $y = 2^{-x}$. Also shown on the diagram are the first 5 of an infinite number of rectangular strips of width 1 unit and height $y = 2^{-x}$ for non-negative integer values of x . For example, the second rectangle shown has width 1 and height $\frac{1}{2}$.



The sum of the areas of the rectangles forms a geometric series.

Show that the limiting sum of this series is 2. **(1 mark)**

b. Show that $\int_0^4 2^{-x} dx = \frac{15}{16 \ln 2}$. **(2 marks)**

c. Use parts (a) and (b) to show that $e^{15} < 2^{32}$. **(2 marks)**

Worked Solutions

1. Calculus, 2ADV C4 2014 HSC 4 MC

$$\begin{aligned} \int e^{2x} dx \\ &= \frac{1}{2} e^{2x} + C \\ &\Rightarrow C \end{aligned}$$

2. Calculus, 2ADV C4 2020 HSC 4 MC

$$\begin{aligned} \int e + e^{3x} &= ex + \frac{1}{3} e^{3x} + c \\ \text{(Note } e \text{ is a simple constant here)} \\ &\Rightarrow B \end{aligned}$$

3. Calculus, 2ADV C4 2012 HSC 9 MC

$$\begin{aligned} \int_1^4 \frac{1}{3x} dx \\ &= \frac{1}{3} [\ln x]_1^4 \\ &= \frac{1}{3} [\ln 4 - \ln 1] \\ &= \frac{1}{3} \ln 4 \\ &\Rightarrow B \end{aligned}$$

TIP: Note that $\ln(1) = 0$, as $e^0 = 1$

4. Calculus, 2ADV C4 2013 HSC 11e

$$\int e^{4x+1} dx = \frac{1}{4} e^{4x+1} + C$$

5. Calculus, 2ADV C4 2004 HSC 3bi

$$\begin{aligned} \int_1^2 e^{3x} dx &= \frac{1}{3} [e^{3x}]_1^2 \\ &= \frac{1}{3} (e^6 - e^3) \end{aligned}$$

6. Calculus, 2ADV C4 2006 HSC 2bi

$$\int 1 + e^{7x} dx = x + \frac{1}{7} e^{7x} + c$$

7. Calculus, 2ADV C4 2015 HSC 11h

$$\begin{aligned}\int \frac{x}{x^2-3} \, dx \\&= \frac{1}{2} \int \frac{2x}{x^2-3} \, dx \\&= \frac{1}{2} \ln(x^2-3) + C\end{aligned}$$

8. Calculus, 2ADV C4 2018 HSC 11e

$$\begin{aligned}\int_0^3 e^{5x} \, dx &= \left[\frac{1}{5} e^{5x} \right]_0^3 \\&= \frac{1}{5} e^{15} - \frac{1}{5} e^0 \\&= \frac{1}{5} (e^{15} - 1)\end{aligned}$$

9. Calculus, 2ADV C4 2016 HSC 12d

i. $y = xe^{3x}$

Using product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot 3e^{3x} + 1 \cdot e^{3x} \\&= e^{3x} (1 + 3x)\end{aligned}$$

ii. $\int_0^2 e^{3x} (3 + 9x) \, dx$

$$\begin{aligned}&= 3 \int_0^2 e^{3x} (1 + 3x) \, dx \\&= 3 \left[xe^{3x} \right]_0^2 \\&= 3 (2e^6 - 0) \\&= 6e^6\end{aligned}$$

10. Calculus, 2ADV C4 2004 HSC 3bii

$$\begin{aligned}\int \frac{x}{x^2-3} \, dx &= \frac{1}{2} \int \frac{2x}{x^2-3} \, dx \\&= \frac{1}{2} \ln(x^2-3) + C\end{aligned}$$

11. Calculus, 2ADV C4 2006 HSC 2bii

$$\begin{aligned}\int_0^3 \frac{8x}{1+x^2} \, dx \\&= 4 \int_0^3 \frac{2x}{1+x^2} \, dx \\&= 4 \left[\log_e (1+x^2) \right]_0^3 \\&= 4 [\log_e (1+9) - \log_e (1+0)] \\&= 4 [\log_e 10 - \log_e 1] \\&= 4 \log_e 10\end{aligned}$$

12. Calculus, 2ADV C4 2019 MET1-N 3

$$\begin{aligned}\int_0^1 e^x - e^{-x} \, dx &= \left[e^x + e^{-x} \right]_0^1 \\&= \left[e + \frac{1}{e} - (1 + 1) \right] \\&= e + \frac{1}{e} - 2\end{aligned}$$

13. Calculus, 2ADV C4 2020 HSC 17

$$\begin{aligned}\int \frac{x}{4+x^2} \, dx &= \frac{1}{2} \int \frac{2x}{4+x^2} \, dx \\&= \frac{1}{2} \log_e (4+x^2) + c\end{aligned}$$

14. Calculus, 2ADV C4 SM-Bank 1

$$\begin{aligned}\int_1^3 \frac{2}{x} \, dx &= [2 \log_e x]_1^3 \\m &= 2 \log_e 3 - 2 \log_e 1 \\&= 2 \log_e 3 \\\therefore e^m &= e^{2 \log_e 3} \\&= e^{\log_e 9} \\&= 9\end{aligned}$$

15. Calculus, 2ADV C4 SM-Bank 3

$$\begin{aligned} f(x) &= \int f'(x) \, dx \\ &= \int \frac{x^2}{x^3 + 1} \, dx \\ &= \frac{1}{3} \int \frac{3x^2}{x^3 + 1} \, dx \\ &= \frac{1}{3} \log_e |(x^3 + 1)| + c \end{aligned}$$

Since $f(1) = \log_e 2$,

$$\begin{aligned} \frac{1}{3} \log_e 2 + c &= \log_e 2 \\ c &= \frac{2}{3} \log_e 2 \end{aligned}$$

$$\therefore f(x) = \frac{1}{3} \log_e (x^3 + 1) + \frac{2}{3} \log_e 2$$

16. Calculus, 2ADV C4 2005 HSC 2ci

$$\begin{aligned} \int \frac{6x^2}{x^3 + 1} \, dx &= 2 \int \frac{3x^2}{x^3 + 1} \, dx \\ &= 2 \ln (x^3 + 1) + C \end{aligned}$$

17. Calculus, 2ADV C4 2008 HSC 2ci

$$\begin{aligned} \int \frac{dx}{x + 5} \\ &= \ln(x + 5) + C \end{aligned}$$

18. Calculus, 2ADV C4 2010 HSC 2dii

$$\begin{aligned} \int \frac{x}{4 + x^2} \, dx \\ &= \frac{1}{2} \int \frac{2x}{4 + x^2} \, dx \\ &= \frac{1}{2} \ln(4 + x^2) + C \end{aligned}$$

IMPORTANT: Minimise errors by adjusting the integral to fit the form $\frac{f'(x)}{f(x)}$ before integrating.

19. Calculus, 2ADV C4 2011 HSC 4b

$$\begin{aligned} \int_e^{e^3} \frac{5}{x} \, dx \\ &= 5 \int_e^{e^3} \frac{1}{x} \, dx \\ &= 5 [\ln x]_e^{e^3} \\ &= 5 (\ln e^3 - \ln e) \\ &= 5(3 - 1) \\ &= 10 \end{aligned}$$

MARKER'S COMMENT: Most common error was $\ln(5x)$. Minimize errors by getting the integral in the form of $\frac{f'(x)}{f(x)}$ before integrating.

20. Calculus, 2ADV C4 2012 HSC 12b

$$\begin{aligned} \int \frac{4x}{x^2 + 6} \, dx \\ &= 2 \int \frac{2x}{x^2 + 6} \, dx \\ &= 2 \ln(x^2 + 6) + C \end{aligned}$$

21. Calculus, 2ADV C4 2013 HSC 11f

$$\begin{aligned} \int_0^1 \frac{x^2}{x^3 + 1} \, dx \\ &= \frac{1}{3} \int_0^1 \frac{3x^2}{x^3 + 1} \, dx \\ &= \frac{1}{3} [\ln(x^3 + 1)]_0^1 \\ &= \frac{1}{3} (\ln 2 - \ln 1) \\ &= \frac{1}{3} \ln 2 \end{aligned}$$

TIP: Note that $\ln(1) = 0$, because $e^0 = 1$

22. Calculus, 2ADV C4 2019 HSC 13c

i. $y = (\ln x)^2$

$$\frac{dy}{dx} = 2 \cdot \frac{1}{x} \cdot \ln x$$
$$= \frac{2\ln x}{x}$$

ii. $\int \frac{\ln x}{x} \, dx = \frac{1}{2} \int \frac{2\ln x}{x} \, dx$

$$= \frac{1}{2} (\ln x)^2 + C$$

♦ Mean mark part (ii) 49%.

23. Calculus, 2ADV C4 2020 HSC 18

a. $y = e^{2x}(2x + 1)$

$$\frac{dy}{dx} = 2e^{2x}(2x + 1) + 2e^{2x}$$
$$= 2e^{2x}(2x + 2)$$
$$= 4e^{2x}(x + 1)$$

b. $\int (x + 1)e^{2x} \, dx = \frac{1}{4} \int 4e^{2x}(x + 1) \, dx$

$$= \frac{1}{4} e^{2x}(2x + 1) + c$$

♦ Mean mark part (b) 40%.

24. Calculus, 2ADV C4 2022 HSC 29

a. Consider the rectangle heights:

$$2^0 = 1, \, 2^{-1} = \frac{1}{2}, \, 2^{-2} = \frac{1}{4}, \, 2^{-3} = \frac{1}{8}, \dots$$
$$\Rightarrow \text{Rectangle Areas} = 1, \, \frac{1}{2}, \, \frac{1}{4}, \, \frac{1}{8}, \dots$$
$$a = 1, \, r = \frac{1}{2}$$
$$S_\infty = \frac{a}{1 - r} = \frac{1}{1 - \frac{1}{2}} = 2 \dots \text{as required}$$

b. Show $\int_0^4 2^{-x} \, dx = \frac{15}{16\ln 2}$

Mean mark (b) 56%.

$$\int_0^4 2^{-x} \, dx = \frac{-1}{\ln 2} [2^{-x}]_0^4$$
$$= \frac{-1}{\ln 2} \left(\frac{1}{16} - 1 \right)$$
$$= \frac{1}{\ln 2} - \frac{1}{16\ln 2}$$
$$= \frac{16 - 1}{16\ln 2}$$
$$= \frac{15}{16\ln 2} \dots \text{as required}$$

c. Show $e^{15} < 2^{32}$

Area under curve < Sum of rectangle areas

♦♦♦ Mean mark (c) 9%.

$$\frac{15}{16\ln 2} < 2$$
$$15 < 32\ln 2$$
$$\frac{15}{32} < \ln 2$$
$$e^{\frac{15}{32}} < e^{\ln 2}$$
$$\sqrt[32]{e^{15}} < 2$$
$$e^{15} < 2^{32} \dots \text{as required}$$